

Optical Path Length (OPL) Drift Modelling and Feedback Control

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1 Introduction

This report presents the formulation and experimental validation of an Optical Path Length (OPL) drift model used to stabilise the EXOhSPEC spectrograph. The model quantifies the influence of temperature, pressure, and humidity on OPL variations and enables feedback-based temperature correction via a Thermo-Electric Cooler (TEC).

2 Mathematical Model

Let the measured OPL deviation at time t be

$$y(t) = \Delta\text{OPL}(t).$$

The predicted deviation is expressed as

$$\hat{y}(t) = a_0 + a_1 t + \sum_{k \in \mathcal{K}} w_k (u_k * g_{\tau_k, \theta_k})(t),$$

where $*$ denotes convolution, w_k are channel sensitivities, and $g_{\tau, \theta}$ is a first-order kernel

$$g_{\tau, \theta}(t) = \frac{1}{\tau} e^{-(t-\theta)/\tau} \mathbf{1}\{t \geq \theta\}.$$

Inputs include temperature, pressure, and humidity perturbations:

$$u_k \in \{\Delta T^+(t), \Delta T^-(t), \Delta P(t), \Delta H(t)\},$$

with $\Delta T^+ = \max(\Delta T, 0)$ and $\Delta T^- = \min(\Delta T, 0)$.

Channel Gains

Because $\int g_{\tau, \theta} dt = 1$, steady-state sensitivities are

$$g_T = \sum_m (w_m^{T^+} + w_m^{T^-}), \quad g_P = \sum_m w_m^P, \quad g_H = \sum_m w_m^H.$$

Each represents OPL sensitivity to the respective environmental parameter.

3 Frozen Model and Error Metrics

For new data, (w_k, τ_k, θ_k) remain fixed while (a_0, a_1) are refitted (“frozen model”). Model accuracy is measured via:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_i (y_i - \hat{y}_i)^2}, \quad \text{MAE} = \frac{1}{N} \sum_i |y_i - \hat{y}_i|, \quad R^2 = 1 - \frac{\sum_i (y_i - \bar{y})^2}{\sum_i (y_i - \hat{y}_i)^2}.$$

Example sensitivities:

$$g_T = -8.22 \times 10^3 \text{ } \mu\text{m } ^\circ\text{C}^{-1}, \quad g_P = 0.53 \text{ } \mu\text{m hPa}^{-1}, \quad g_H = -0.21 \text{ } \mu\text{m } \%^{-1}.$$

4 Feedback Control Law

To counter OPL drift, the commanded temperature correction is

$$\Delta T_{\text{cmd}} = -\frac{\Delta \text{OPL} - g_P \Delta P - g_H \Delta H}{g_T}.$$

The controller is implemented as a PI loop with anti-windup and a 900 s thermal lag allowance.

5 Feedback Loop Evaluation

Three feedback experiments (Exp-A, Exp-B, Exp-C) were performed 3–5 Nov 2025 with progressively refined algorithms.

Table 1: Feedback-phase metrics. β_Y is pixel–temperature sensitivity.

Experiment	Date	Type	dY_{rms} (px)	β_Y (px/ $^\circ\text{C}$)	R^2	$\Sigma du $ (mK)
Exp-A	2025-11-03	Ultrafine	8.53	+167.4	0.11	2.1
Exp-B	2025-11-05	Hybrid #1	4.88	+1.32	0.05	1000
Exp-C	2025-11-05	Hybrid #2	3.06	56.6	0.77	112

Interpretation. β_Y represents the change in pixel position per degree Celsius:

$$dY = \beta_Y \Delta T + \varepsilon.$$

A consistent β_Y and high R^2 imply deterministic thermal coupling. Exp-C exhibited $\beta_Y = -56.6 \text{ px}/^\circ\text{C}$ ($\approx 0.018 \text{ } ^\circ\text{C}$ per px), confirming stable coupling.

6 Performance Evolution

- **Exp-A (Ultrafine):** Open-loop; random pixel drift uncorrelated with temperature.
- **Exp-B (Hybrid #1):** Thermal feedback reduced mean drift 45 %, but sign flips occurred due to missing cooldown logic.
- **Exp-C (Hybrid #2):** Stable negative β_Y ; dY_{rms} reduced by 64 % relative to Exp-A.

7 Model and Figures

7.1 $\Delta X/\Delta Y$ and Temperature Evolution

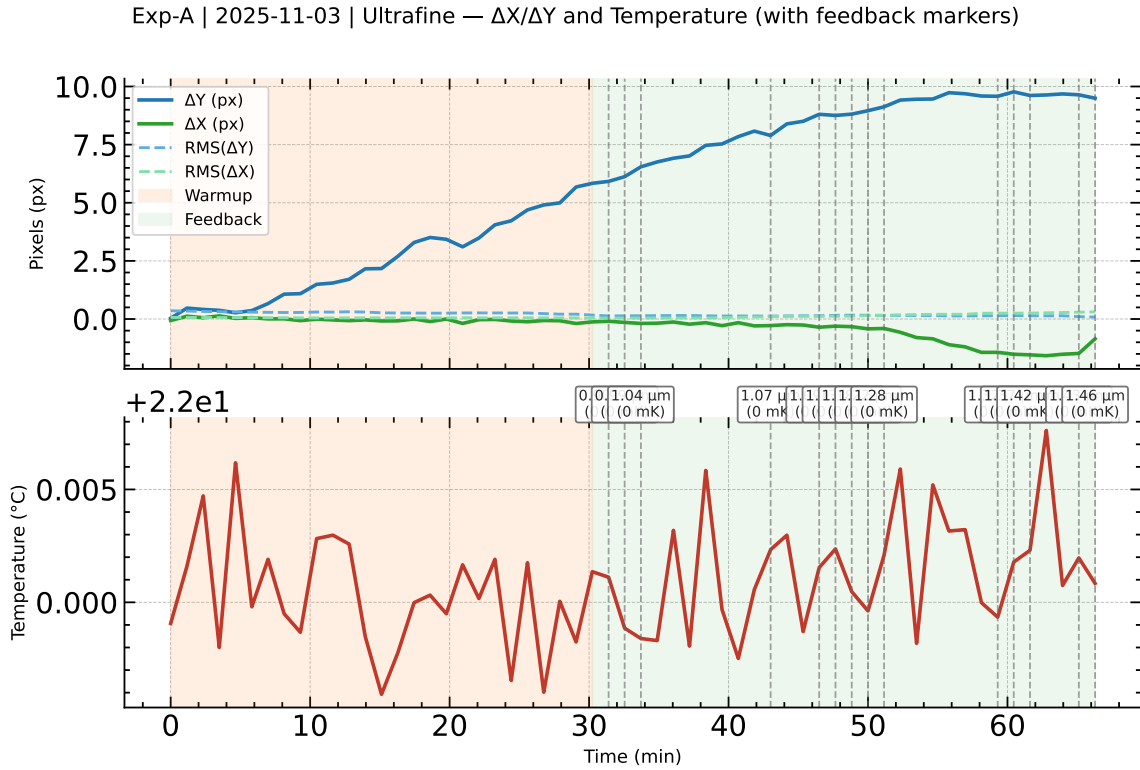


Fig. 1: Experiment A (Ultrafine, 2025-11-03): ΔX , ΔY , and temperature. Orange = warm-up; green = feedback. Vertical dashed lines show feedback actions with equivalent OPL change (μm) and temperature step (mK).

Exp-B | 2025-11-05 | Hybrid #1 — $\Delta X/\Delta Y$ and Temperature (with feedback markers)

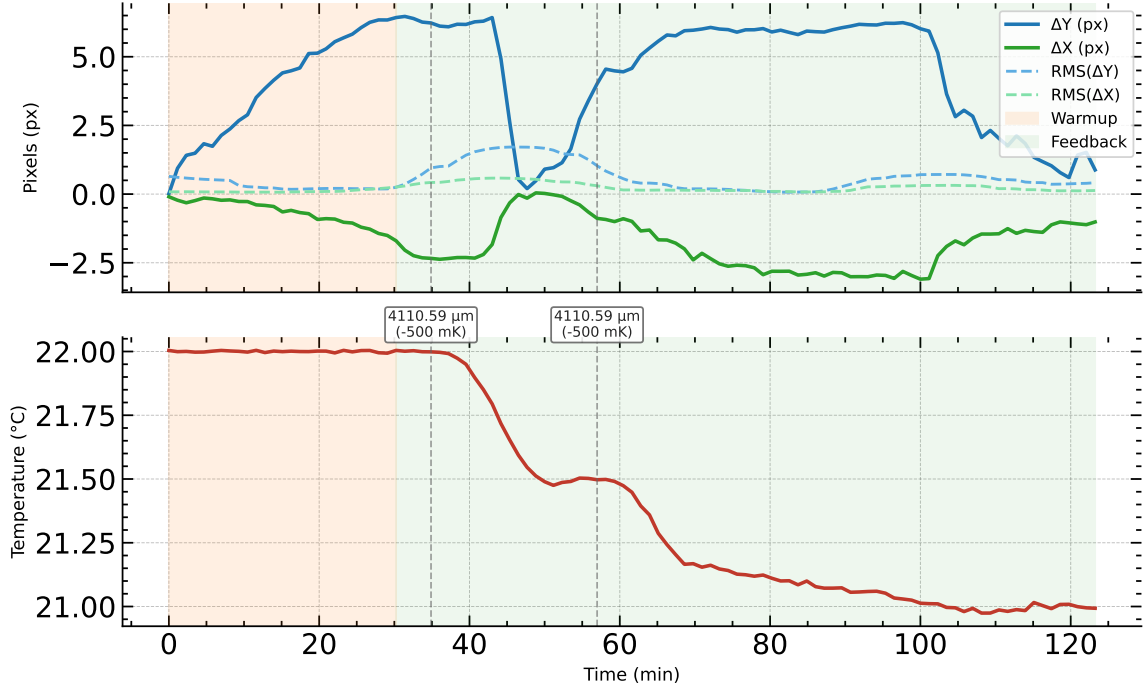


Fig. 2: Experiment B (Hybrid #1, 2025-11-05): Pixel drift and temperature. Dashed RMS traces show short-term jitter; weak correlation indicates delayed thermal response.

Exp-C | 2025-11-05 | Hybrid #2 — $\Delta X/\Delta Y$ and Temperature (with feedback markers)

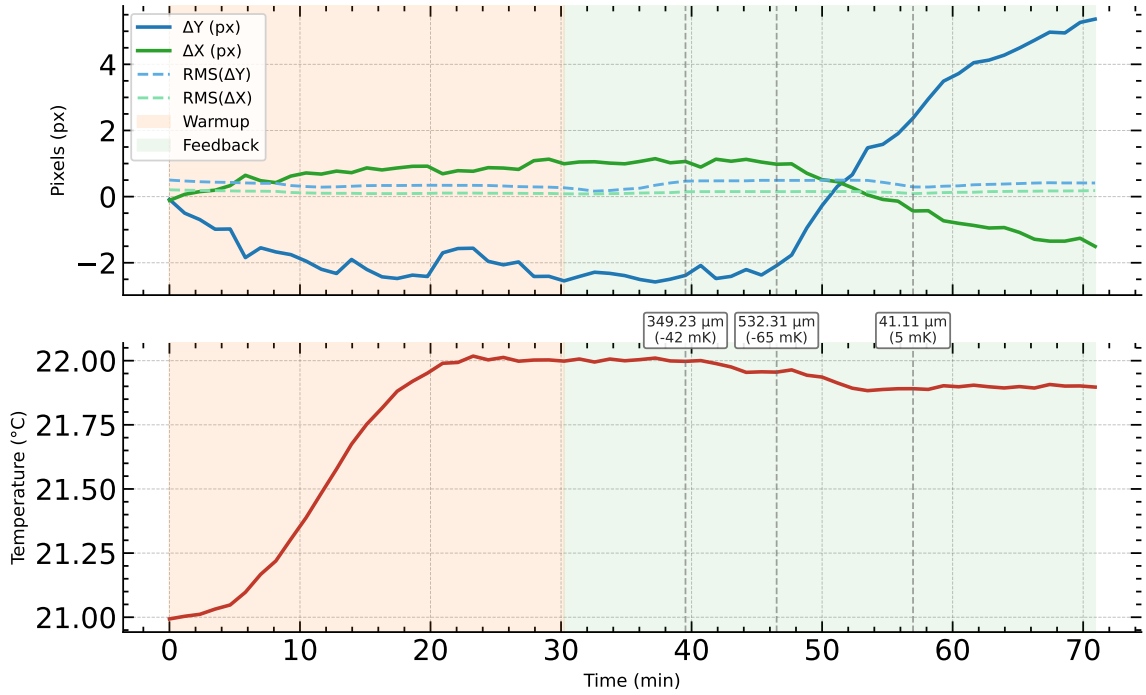


Fig. 3: Experiment C (Hybrid #2, 2025-11-05): Stable negative β_Y . Cooling steps (42 mK, 65 mK) produced the expected +Y excursions.

7.2 OPL Residuals and Environmental Parameters

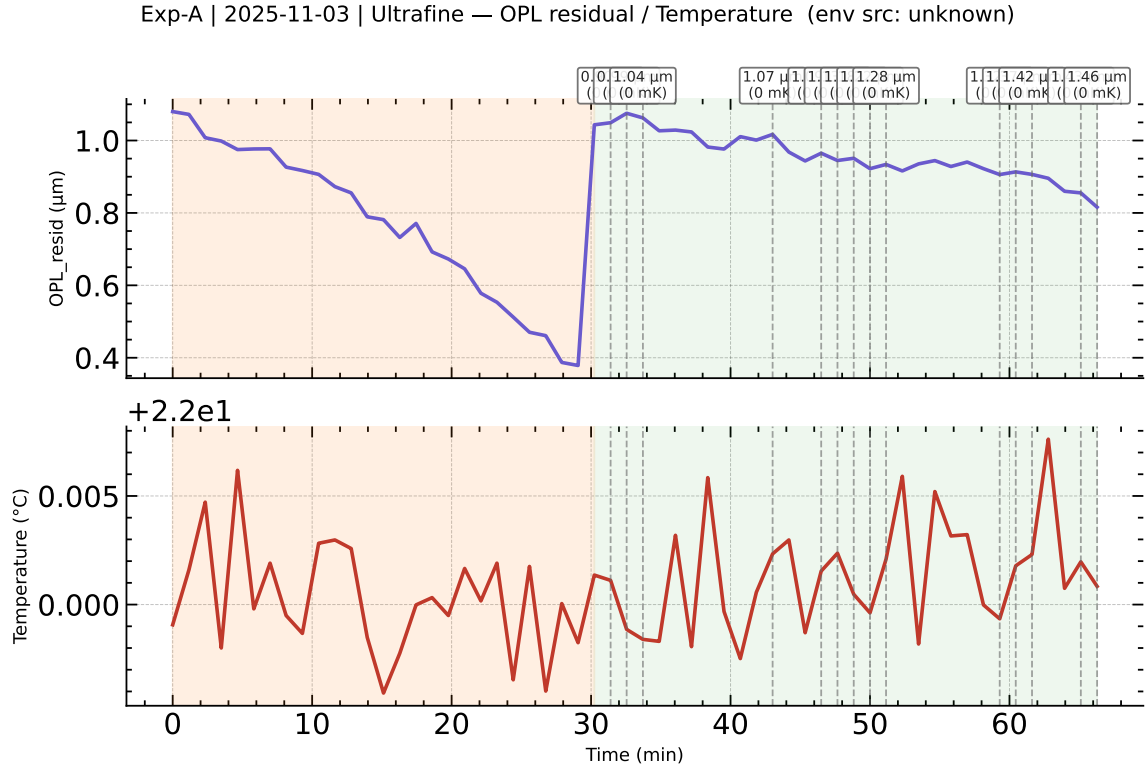


Fig. 4: Exp-A: OPL residual and temperature. No pressure/humidity data available (environment source unknown). $OPL_{\text{resid}} = OPL_{\text{meas}} - (a_0 + a_1 t) - g_P \Delta P - g_H \Delta H$.

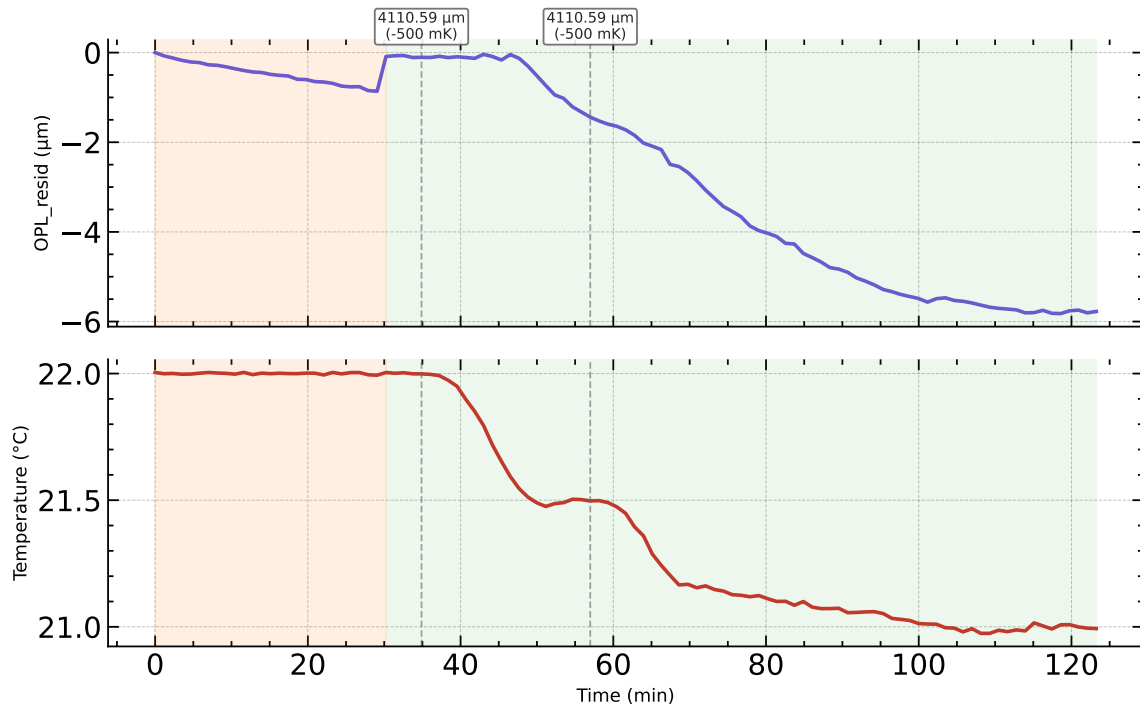


Fig. 5: Exp-B: OPL residual and temperature with partial ΔP , ΔH records. Residual correlation with pressure/humidity was minor during feedback.

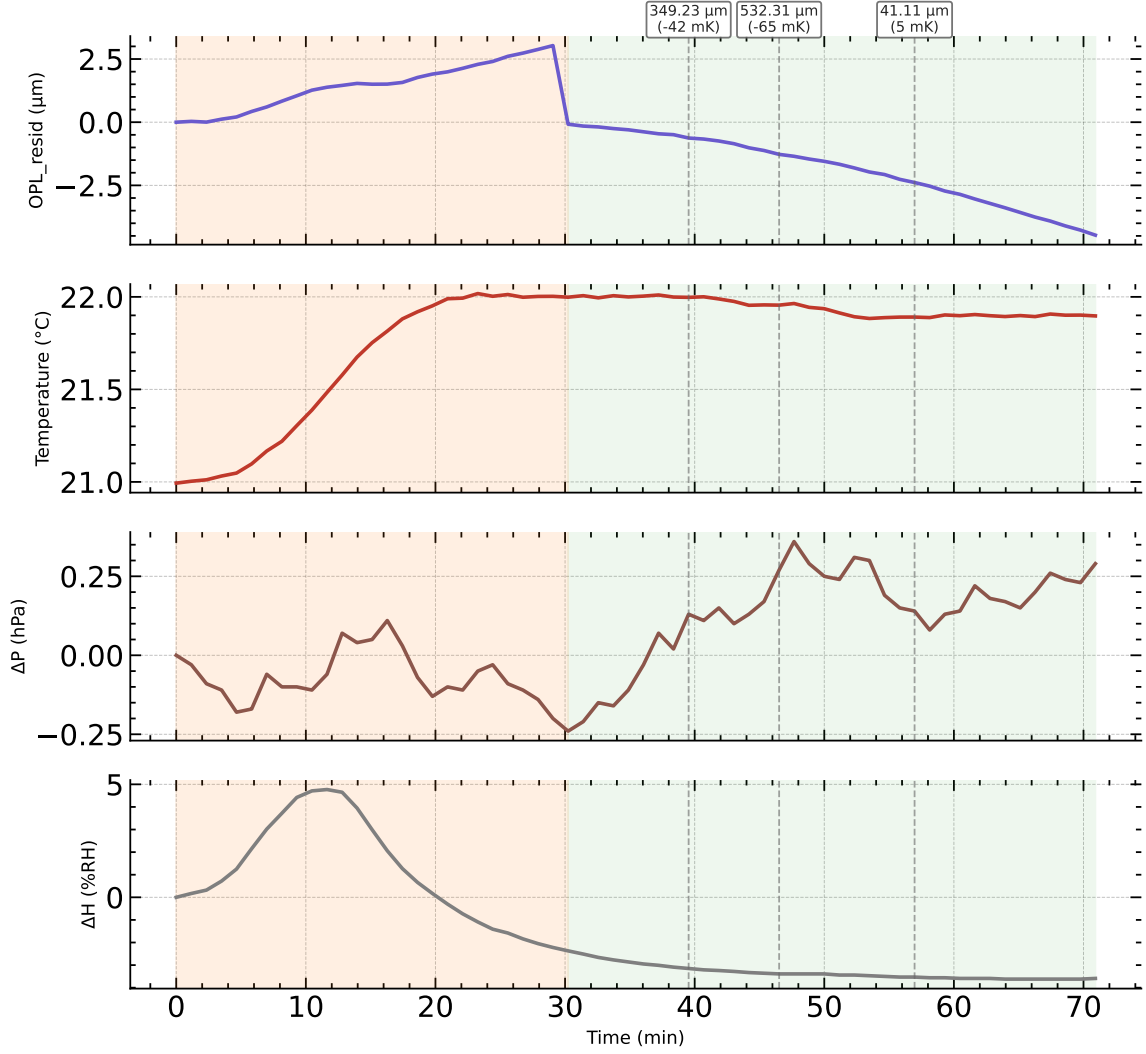


Fig. 6: Exp-C: OPL residual, temperature, and ECU-measured pressure/humidity. Pressure variations contributed $< 0.5 \mu\text{m}$ to total OPL drift.

8 Refined Feedback Model

8.1 Hybrid Controller Formulation

$$\text{OPL}_{\text{resid}} = \text{OPL} - \widehat{\text{OPL}}(t) - g_P \Delta P - g_H \Delta H, \quad \widehat{\text{OPL}}(t) = a_0 + a_1 t.$$

Equivalent temperature errors:

$$e_T^{\text{OPL}} = -\frac{\text{OPL}_{\text{resid}}}{g_T}, \quad e_T^{\text{PX}} = -\frac{d_{\text{dom}}}{\beta}, \quad d_{\text{dom}} = \arg \max(|\Delta X|, |\Delta Y|).$$

Weighted fusion:

$$e_T = w e_T^{\text{OPL}} + (1 - w) e_T^{\text{PX}}, \quad w \approx 0.85.$$

PI control:

$$u_{\text{PI}} = K_p e_T + K_i \int e_T dt, \quad \Delta T = \text{clip}(u_{\text{PI}}, \Delta T_{\min}, \Delta T_{\max}).$$

9 Adaptive Refinement Strategy

9.1 Limitations

Fixed-gain PI control cannot compensate for variable β_Y or nonlinear thermal dynamics; large corrections need 5–8 min cooldowns to avoid overshoot.

9.2 Model-Based Fusion Controller

At each update, choose ΔT^* to minimise

$$J(\Delta T) = w_1(\text{OPL}_{\text{resid}} + g_T \Delta T)^2 + w_2(r_{\text{px}} + \beta \Delta T)^2 + \lambda \Delta T^2,$$

yielding

$$\Delta T^* = -\frac{w_1 g_T \text{OPL}_{\text{resid}} + w_2 \beta r_{\text{px}}}{w_1 g_T^2 + w_2 \beta^2 + \lambda}.$$

9.3 Adaptive β and Scheduling

Recursive least-squares updates β_X, β_Y every few minutes (forgetting 0.98). Sign-lock after $R^2 > 0.6$ for 10 min; large steps ($|\Delta T| > 0.15^\circ\text{C}$) trigger 5–8 min holds.

10 Three-Phase Refinement Roadmap

Phase 1 – Metrology and Calibration

- Reduce centroid noise to 0.03 px using Gaussian fits and averaging.
- Re-measure $g_T, g_P, g_H, \beta_{X,Y}$ under current alignment.
- Suppress modal noise via slow fibre/diffuser motion.

Phase 2 – Adaptive Model-Based Control

- Implement optimal ΔT^* from Eq. (above).
- Update weights w_1, w_2 from rolling variances.

- Enable adaptive β with sign-lock and variance tracking.
- Dual-rate updates: 5–10 min micro-steps ($|\Delta T| < 30$ mK), 5–8 min cooldowns after macro-steps.

Phase 3 – Verification and Tightening

- Monitor dY_{rms} , OPL residuals, and β_Y stability.
- Correlate ΔT with subsequent ΔdY to confirm causality.
- Target: $dY_{\text{rms}} < 1.5$ px (initial) $\rightarrow$ 0.1 px (stable) $\rightarrow$ 0.01 px effective.

11 Expected Outcome

The adaptive fusion controller integrates OPL, pixel, and environmental feedback into a self-calibrating architecture. By dynamically weighting sensor reliability and enforcing thermal scheduling, the system can reach sub-0.01 px image stability—equivalent to milli-Kelvin-level temperature control in the spectrograph enclosure.